Forest-to-energy in a mountain village, and where a CTL trial could fit

Project overview for Komatsu Ltd.
Forest and Agriculture Business Division



Teams meeting · Tuesday 27 October 2026 · 14:00–14:45 JST

Rob Oudendijk · Mitsue-it (independent initiative)

A mountain village with a forest that needs managing

≈ 1,400
residents

≈ 93 %
forest cover

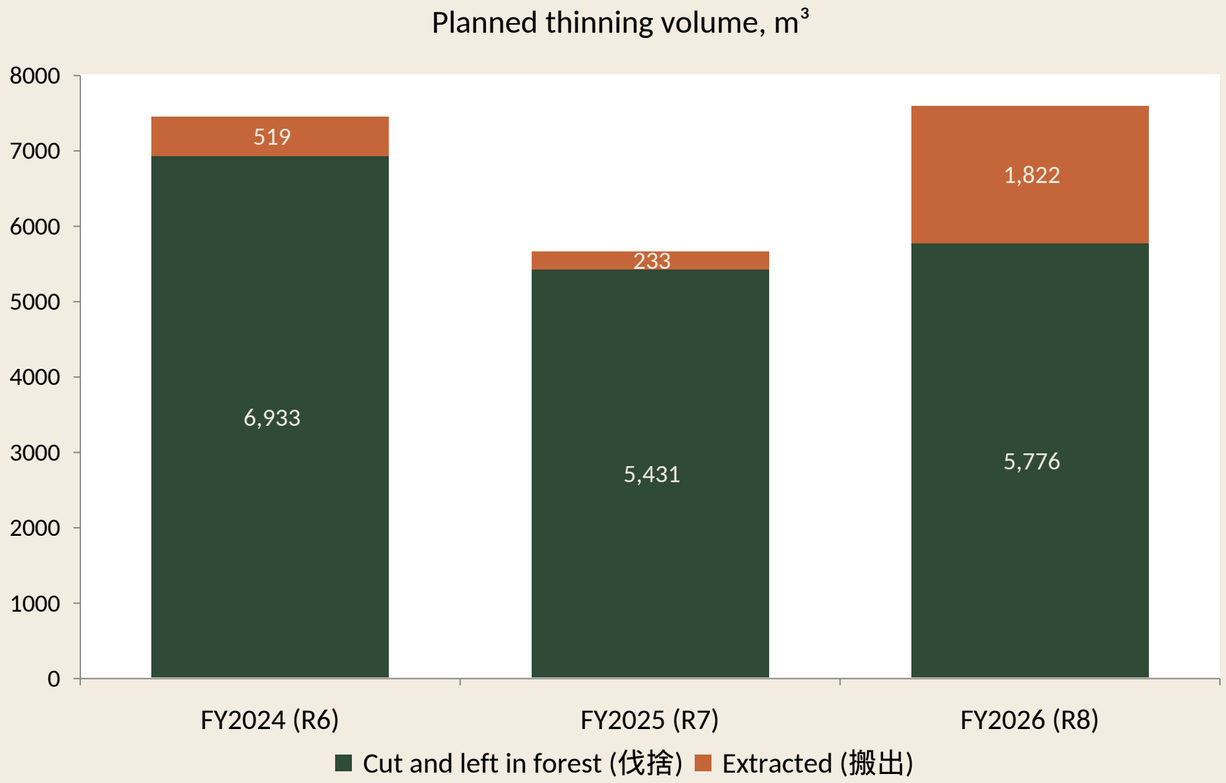
50–75 yr
age of many unmanaged sugi plantations

Mitsue is a village in Uda district, Nara Prefecture. Much of its sugi (cedar) was planted after the war and has not been managed for decades.

Forestry labour is the bottleneck, not land or demand. The cooperative is small and has few young people, so we need machines and new operators.

An independent initiative working with the village, the local forestry cooperative and the NGO more trees. Nothing is contracted at this stage.

We want to harvest, but we do not have enough people



≈ 20,700 m³

planned thinning, FY2024–26: 113 parcels, ≈ 222 ha in 4 districts

≈ 88 % is cut and left

Only ≈ 2,600 m³ is planned to be extracted.

The cooperative's crew is small and has few young people. Labour, plus slope and access, keeps the wood in the forest.

Our reading of the plan; to be confirmed with the cooperative.

Source: village hall thinning plan R6–R8 (plans, not realised harvests).

Use the wood as fuel, then replant native broadleaf

1 Thin and harvest

Cut-to-length machines on steep sugi stands

2 Chip and dry

Local chipping, drying with CHP heat

3 50 kWe CHP

Small gasification unit, 24 h operation

4 Power and heat

For local use; a small compute load later

Replanting

Cleared stands go back to native broadleaf (oak, chestnut, konara), chosen with the NGO more trees. The village's default is native broadleaf, not another monoculture.

Fuel demand, first unit

A 50 kWe unit running 24 h needs ≈ 1.3–1.8 t dry wood per day, about 450–600 t a year. Three estimating methods agree on that range. No unit has been chosen yet.

A 50 kWe prototype at a local forestry site

Probable site: a local forestry cooperative's yard, using a 20 ft container. The cooperative is keen on the CHP. Not final yet.

Why there: wood arrives there anyway, so fuel handling and drying are simple.

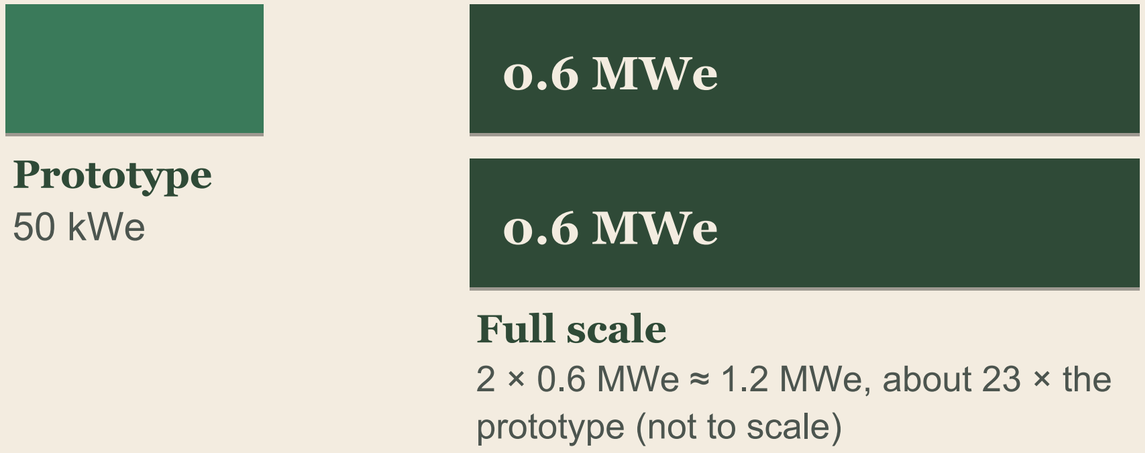
What we do first: analyse the fuel, choose a unit, agree the site and the wood flow with the cooperative and village hall.

Why it matters to CTL: the prototype creates the steady demand that makes extracting wood worthwhile.

Photos of the planned site

to be added once the cooperative sends them

Where it leads: ≈ 1.2 MWe of CHP and a full-scale AI data center



Prototype
50 kWe

Full scale
2 × 0.6 MWe ≈ 1.2 MWe, about 23 × the prototype (not to scale)

Two units, not one, so the plant keeps running during maintenance and the output stays modular.

Roughly 8,000 MWh of power a year, and a similar scale of heat for drying the fuel.

What it takes

A full-scale AI data center as the anchor customer: steady 24-hour power and heat from dispatchable biomass.

About 13,000 t of dry wood a year, which needs a doubled forest crew and mechanised thinning and harvest.

Long-term, with no fixed date. The prototype comes first.

Mitsue is the first. Many more forests need the same

≈ 10 million ha

of planted forest in Japan.

Much of it was planted after the war and is now ready to harvest, in villages that, like Mitsue, are short of forestry workers.

Source: Forestry Agency, 31 March 2022 (10,093,422 ha).

1 Mitsue: prove it

Harvest, replant native broadleaf, run a small CHP, train local operators.

2 Nearby villages: repeat it

Same machines, same methods, same open documents. We have had early conversations in other forest villages.

3 Many more forests

A template for depopulating mountain villages across Japan, with CTL machines and trained operators at the centre.

What we hope to learn from a Komatsu trial

Our situation

Steep terrain, thinning of 50–75-year-old sugi.

Not enough workers: the cooperative has few young people.

Small-diameter thinning, with some larger stems in long-unmanaged stands.

What we have looked at

901XC with C93 head looks like the better match for our scale.

931XC and 855 forwarder also considered.

Impression from videos only; we have not seen the machines in the field.

Questions for Komatsu

Is a trial feasible at our site?

Slope limits and extraction-trail spacing?

Is permanent road needed, or only light forwarder trails?

Which model and head suit our stands?

What training could a local trainee operator get?

Machines only help if there are people to run them

The gap

The cooperative's crew is small and has few young people.

Thinning and harvest at the scale the forest needs is not possible with today's team.

A cut-to-length machine lets a small crew do much more.

A local successor

A young woman who works at the cooperative, bright and determined, and eager to learn to operate machines like the 931XC.

We would like her to train on Komatsu machines if a trial goes ahead.

Why it matters to Komatsu

A local career path for young people in a depopulating village, on Japanese-made machines.

A chance to bring more women into forestry machine work and widen the workforce.

A story that is easy to tell: a trial that creates the next operator.

What training, licence and machine access are possible for a trainee?

Early planning, with a realistic trial window around early 2028



Phase 0

Local trust, site and fuel talks

Prototype

50 kWe CHP at a local forestry site

Phase 1

Small ≈ 50 kW compute load; site undecided

CTL trial

End of FY2027 is our realistic window

Full scale

≈ 1.2 MWe CHP and a full-scale AI data center; no fixed date

Status: no contracts, purchases or funding are committed. The dates are targets, not promises, and the full-scale facility has no fixed date.

Three ways we could work with Komatsu

1 Trial first

Mitsue as an additional trial or training site for the CTL programme.

We would share observations on slope, access and extraction, and help train a local operator.

2 Purchase or lease later

Only if the trial shows a good fit and financing is in place.

Neither is committed today.

3 A shared story

A concrete case of Japanese-made forestry machinery supporting broadleaf reforestation, local biomass energy and village revitalisation.

Case studies, joint publicity, hosting visitors, at whatever level suits Komatsu.

And a template for many more villages.

What we would like from today

1 Your view: is a trial feasible, and what would you need to see from us?

2 A later site visit to the village's forests with local forestry operators, once we have something to show on the ground.

3 The right Komatsu contacts for model selection and slope and trail guidance.

4 Training options for a young woman who wants to learn to operate the machines, and how to bring more women into forestry work.

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tinyurl.com/2xv524fz (project repository)

Thank you · ありがとうございます